

Problem 13 – Smart Hotel Reservation System

Difficulty: ★ ★ ★ Beginner+

1. Overview

A hotel wants to replace its handwritten reservation book with a digital reservation system.

Guests should be able to:

- View available rooms.
- Choose a room type.
- Select the number of nights.
- Select a check-in date.
- Add optional breakfast.
- Make a reservation.
- View their reservation summary.
- Cancel a reservation.

The hotel should also be able to track reservations and generate a management dashboard.

 **Streamlit only:** Do not use `input()`, `print()`, or `while True`.

2. Hotel Rooms

Create dictionaries to store the room information.

Room Prices

Room Type	Price Per Night
Standard	1200 EGP
Deluxe	1800 EGP
Family	2500 EGP
Suite	4000 EGP

Available Rooms

Room Type	Available Rooms
Standard	10
Deluxe	6
Family	4
Suite	2

Initially, all rooms must be available.

The number of available rooms must decrease when a reservation is made and increase when a reservation is cancelled.

3. Application Layout

Your Streamlit application must contain:

Sidebar

Display:

- Hotel facilities
- Check-in time
- Check-out time
- Hotel information

The complete room availability list belongs in the **Available Rooms tab**, not the sidebar.

Four Tabs

1. **Make Reservation**
2. **Available Rooms**
3. **Cancel Reservation**
4. **Reservation Summary**

Use `st.tabs()`.

4. Make Reservation Tab

Create an `st.form()` containing all reservation controls.

Guest Name

Use `st.text_input()`.

Room Type

Use `st.selectbox()` to choose a room type.

Number of Nights

Use `st.number_input()`.

The number of nights must be greater than zero.

Check-in Date

Use `st.date_input()`.

Breakfast

Use `st.selectbox()` with:

- Yes
- No

Breakfast costs **250 EGP per night**.

Reserve Room

Use `st.form_submit_button()`.

5. Reservation Validation

When the guest submits the reservation:

- Guest name must not be empty.
- The selected room type must exist.
- At least one room of the selected type must be available.
- Number of nights must be greater than zero.
- A check-in date must be selected.

If the information is invalid, do not create the reservation.

Use `st.error()` for invalid information.

If only one room of the selected type remains, you may display a warning using `st.warning()`.

If no room is available, display an appropriate error or warning message.

6. Calculate Reservation Cost

Calculate the accommodation cost using:

Room Cost = Price Per Night × Number of Nights

If breakfast is selected:

Breakfast Cost = 250 × Number of Nights

The final cost is:

Final Cost = Room Cost + Breakfast Cost

Example

For a Deluxe room:

- Price = 1800 EGP per night
- Nights = 3
- Breakfast = 250 EGP per night

Room cost:

$$1800 \times 3 = 5400 \text{ EGP}$$

Breakfast:

$$250 \times 3 = 750 \text{ EGP}$$

Final cost:

$$5400 + 750 = 6150 \text{ EGP}$$

7. Confirm the Reservation

After a valid reservation:

1. Calculate the room cost.
2. Calculate the breakfast cost if selected.
3. Calculate the final cost.
4. Decrease the available room count by 1.
5. Store the reservation.
6. Display a confirmation message.
7. Make the reservation available in the Reservation Summary tab.

Use:

- `st.spinner()` to simulate reservation processing.
- `st.success()` after a successful reservation.

Example:

 Processing your reservation...

Then:

 Reservation confirmed successfully!

8. Available Rooms Tab

Display all room types with:

- Room type
- Price per night
- Number of available rooms

Example:

Room Type	Price/Night	Available
Standard	1200 EGP	10
Deluxe	1800 EGP	5
Family	2500 EGP	4
Suite	4000 EGP	2

The available room count must update automatically when rooms are reserved or cancelled.

9. Cancel Reservation Tab

Create a section for cancelling an existing reservation.

The guest or receptionist should provide:

Guest Name

Use `st.text_input()`.

Room Type

Use `st.selectbox()`.

Cancel Reservation

Use `st.button()`.

When the button is pressed:

- Check whether the reservation exists.
- If it exists, cancel it.
- Increase the available room count by 1.
- Update the reservation status.
- Display a cancellation confirmation.

Use:

- `st.success()` for a successful cancellation.
- `st.warning()` when the reservation cannot be cancelled.
- `st.error()` for invalid information.

10. Reservation Summary Tab

Display the **most recently created reservation**.

If there is no reservation yet, display an information message using `st.info()`.

Use `st.columns(2)`.

Left Column – Reservation Information

- Guest name
- Room type
- Number of nights
- Check-in date

Right Column – Cost Information

Display:

- Price per night
- Room cost
- Breakfast status
- Breakfast cost
- Final total
- Reservation status

11. Reservation Summary

Use `st.code()` to display a clear formatted reservation summary.

Example:

```
=====
Reservation Summary
=====
Guest : Sara
Room : Deluxe
Check-in : 15/07/2026
Nights : 3
Breakfast : Included
Final Cost : 6150 EGP
Reservation Status : Confirmed
=====
```

12. Required Text Functions

Your application must use **all six**:

Function	Purpose
<code>st.title()</code>	Application title
<code>st.header()</code>	Major sections
<code>st.subheader()</code>	Smaller sections
<code>st.write()</code>	General information
<code>st.code()</code>	Reservation summary
<code>st.divider()</code>	Separate sections

13. Required Input Widgets

Your application must use:

Widget	Purpose
<code>st.text_input()</code>	Guest name
<code>st.selectbox()</code>	Room type
<code>st.number_input()</code>	Number of nights
<code>st.date_input()</code>	Check-in date
<code>st.selectbox()</code>	Breakfast Yes/No
<code>st.form_submit_button()</code>	Make reservation
<code>st.button()</code>	Cancel reservation

All reservation controls must be inside an `st.form()`.

The cancellation action should use a separate `st.button()` in the Cancel Reservation tab.

14. Required Status Elements

You must use **all five**:

Function	Purpose
st.info()	Hotel information / no reservation
st.success()	Reservation or cancellation confirmed
st.warning()	Limited availability / cancellation issue
st.error()	Invalid information / unavailable room
st.spinner()	Simulate reservation processing

15. Store Reservations

Store every reservation in a **list of dictionaries**.

Each reservation should contain information such as:

- Guest name
- Room type
- Number of nights
- Check-in date
- Price per night
- Breakfast selection
- Breakfast cost
- Final cost
- Reservation status

Because Streamlit reruns the application when users interact with it, use `st.session_state` to store:

- Reservation records
- Room availability
- The most recent reservation

The information must remain available when the user:

- Changes tabs.
 - Makes another reservation.
 - Cancels a reservation.
-

16. Required Functions

Create at least these functions:

display_rooms()

Displays room types, prices, and availability.

calculate_reservation_cost()

Calculates:

- Room cost
- Breakfast cost
- Final cost

make_reservation()

Handles the reservation process and updates room availability.

cancel_reservation()

Handles cancellation and updates room availability.

display_reservation_summary()

Displays the latest reservation and its cost.

You may create additional functions if needed.

17. Challenge Mission ★★

Hotel Management Dashboard

Create a **Hotel Management Dashboard** using the stored reservation data.

Display:

Total Reservations

The number of successful reservations.

Total Revenue

The total value of all confirmed reservations.

Most Reserved Room Type

The room type that has been reserved the greatest number of times.

Total Occupied Rooms

The number of rooms currently occupied based on the room availability.

Total Available Rooms

The number of rooms currently available.

Average Stay Duration

Calculate the average number of nights per reservation.

Guest With the Highest Bill

Find the guest associated with the highest reservation bill.

18. Required Streamlit Topics

Displaying Text

- `st.title()`
- `st.header()`
- `st.subheader()`
- `st.write()`
- `st.code()`
- `st.divider()`

Input

- `st.text_input()`
- `st.selectbox()`
- `st.number_input()`
- `st.date_input()`
- `st.form()`
- `st.form_submit_button()`
- `st.button()`

Status

- `st.info()`
- `st.success()`
- `st.warning()`
- `st.error()`
- `st.spinner()`

Layout

- `st.sidebar`
- `st.tabs()`
- `st.columns()`

State

- `st.session_state`
-

19. Python Topics

This project practices:

- Variables
- Strings
- Lists
- Dictionaries
- Integers and floats
- Arithmetic operators
- Comparison operators
- Assignment operators
- Membership operators
- if / elif / else
- for loops
- User-defined functions
- Dictionary methods
- List operations
- Date handling

20. Final Checklist

Before submitting, make sure you have:

Data

- Room price dictionary
- Room availability dictionary
- Reservation records
- Data stored using `st.session_state`

Sidebar

- Hotel facilities
- Check-in time
- Check-out time
- Hotel information

Tabs

- Make Reservation
- Available Rooms
- Cancel Reservation

- Reservation Summary

Reservation Form

- Guest name
- Room type
- Number of nights
- Check-in date
- Breakfast option
- Reserve button

Reservation System

- Input validation
- Room availability check
- Room cost calculation
- Optional breakfast
- Room availability decreases after reservation
- Room availability increases after cancellation
- Reservation summary
- Reservation status

Required Streamlit Components

- All required text functions
- All required input widgets
- All five status elements
- Sidebar
- Four tabs
- Form
- Two-column summary
- Session state

Challenge

- Total reservations
- Total revenue
- Most reserved room type
- Total occupied rooms
- Total available rooms
- Average stay duration

- Guest with highest bill
-

21. Important Restrictions 🚫

This project must be built as a **Streamlit application**.

Do **not** use:

- `input()`
- `print()`
- `while True`
- Console menus

Streamlit handles user interaction and application reruns automatically.

22. Learning Objectives

After completing this project, you should be able to:

- Manage room inventory using dictionaries.
- Track room availability as reservations are created and cancelled.
- Store multiple reservation records.
- Validate reservation information.
- Work with dates.
- Calculate accommodation costs and optional services.
- Update data based on user actions.
- Use `st.session_state` to maintain application data.
- Build a multi-tab Streamlit reservation system.
- Use forms, sidebars, columns, and status messages.
- Generate a basic hotel management dashboard.